MIL-T-49056A (CR)
1 September 1983
Superceding
MIL-T-49056(EL)
1 May 1975

MILITARY SPECIFICATION

# TEST SET, TRANSPONDER SET AN/APM-305A

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE.

1.1 This specification covers one type test set which is used in conjunction with the AN/USM-281 Oscilloscope, or equivalent, and the AN/APM-239 Transponder Bench Test Set to check, maintain, and calibrate IFF (Identification-Friend-or-Foe) Transponders. The test set is designated Test Set, Transponder Set AN/APM-305A.

## 2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-T-76 | Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing) |
| PPP-P-291 | Paperboard, wrapping, Cushioning |
| PPP-F-320 | Fiberboard, Corrugated and solid, Sheet Stock (Container Grade), and Cut Shapes |
| PPP-B-585 | Box, Wood, Wirebound |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock Corner |
| PPP-B-636 | Box, Fiberboard |
| QQ-S-781 | Strapping, Steel, Flat and seals |

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|---|---|
| MIL-P-116 | Preservation, Packaging Methods of |
| MIL-B-117 | Bag, Sleeve and Tubing-Interior Packaging |
| MIL-P-11268 | Parts, Materials, and Processes used in Electronic Equipment |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this Document or by letter.

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| | |
|---|---|
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finish for Ground Electronic Equipment |
| MIL-P-19644 | Plastic Molding Material (Polystyrene Foam Expanded Bead) |
| MIL-T-21200 | Test Equipment for use with electronic and Electrical Equipment, General Specification for |
| MIL-P-26514 | Polyurethane Foam, Rigid or Plastic, for Packaging |
| MIL-P-55110 | Printed Wiring Boards |
| MIL-C-55442 | Cable Assemblies and Cord Assemblies, Packaging of |

DOD

| | |
|---|---|
| DOD-AIMS 65-1000 (Classified-CONFIDENTIAL) | Technical Standard for the ATCRBS/IFF/Mark XII System |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables of Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electrical Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics, Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics Measurement of |
| MIL-STD-781 | Reliability Tests Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods |

U.S. NATIONAL AVIATION

| | |
|---|---|
| 1010.51A | FAA Mark X(SIF) Air Traffic Control Radar Beacon System (ATCRBS) Characteristics |

INSTRUCTIONS

| | |
|---|---|
| SPI 1G00119 | Special Packaging Instructions |

(Copies of specifications, standards, drawings,and publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or a directed by the Contracting Officer).

3. REQUIREMENTS.



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3.1 DESCRIPTION. The Test Set, Transponder Set AN/APM-305A consists of a Test Set, Transponder TS-3395/APM-305A with a removable cover and associated cable assemblies, accessory attenuators, connector adaptors, terminations, and conversion chart. The TS-3395/APM-305A provides stimuli and measurement capabilities required for shop maintenance of IFF transponders. The test set checks operational characteristics, such as receiver bandpass characteristics and sensitivity, decoding and coding of all modes, transmitter output power and frequency, and sidelobe suppression. The TS-3395/APM-305A operates from an external 115 VAC power source.

3.2 FIRST ARTICLE. When specified in the contract, the contractor shall furnish first article units (See 4.3).

3.3 GENERAL REQUIREMENTS. The AN/APM-305A shall be in accordance with MIL-T-21200, Class 3, unless otherwise specified herein (See 4.4).

3.4 PARTS, MATERIALS AND PROCESSES. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4.1 GENERAL.

a. When non-pressurized castings are used, the requirement for impregnation as stated in MIL-STD-276 shall not apply.

b. When anti-sieze compounds may affect RF grounding, the requirement for their use shall not apply.

3.5 FINISH, PROTECTIVE. The AN/APM-305A shall be given protective finish in accordance with Specification MIL-F-14072. This includes primer and finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be final Film E. semigloss, Light-gray enamel, conforming to Specification MIL-F-14072 (See 4.4).

3.6 MARKING. (See 4.4)

3.6.1 GENERAL. Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group I, as described in that specification (See 4.4).

3.6.2 VISIBILITY. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment (See 4.4).

3.6.3 SERIAL NUMBERS. The following items shall have serial numbers:

a. Test Set, Transponder Set AN/APM-305A.

b. Test Set, Transponder TS-3395/APM-305A (See 4.4)



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3.7 FRONT PANEL PROTECTION. Handles shall be provided on the front of the TS-3395/APM-305A hereafter known as the equipment. The outer edge of the handles shall extend at least the distance of the controls, indicator or electrical connector indicator or electrical connector that has the greatest protrusion from the front panel (See 4.6).

3.8 INPUT POWER SOURCE. The equipment shall be capable of operating from a 115 VAC, ±10%, 60/400 Hz power source in accordance with MIL-T-21200, except the low frequency limit for the power source shall be 45 Hz (See 4.7). The 400 $H_Z$ adapter defined in specification MIL-T-21200 shall not be supplied.

3.8.1 POWER SWITCH. A two position toggle switch, identified as "Power, ON-OFF", shall permit the equipment to be energized by the 115 VAC power source (See 4.7.1).

3.8.2 PROTECTION, ELECTRICAL. Electrical overload protection shall be in accordance with Requirment 8 of MIL-STD-454. Fuses shall be of the indicating type and shall be in accordance with Requirement 39 of MIL-STD-454.

3.8.3 POWER INDICATOR. AN AC power indicator lamp and lamp holder with lens, shall be provided on the front panel of the equipment, to indicate the application of the 115 VAC power source. The indication shall be green in color (See 4.7.2).

3.9 EXTENDER CARD. An extender card shall be provided for the equipment printed circuit cards and shall be keyed to prevent improper insertion of itself or the circuit cards. The extender card shall be stored with the printed circuit cards (See 4.8).

3.10 INSTRUMENT CASE. An instrument case shall be provided to house the equipment chassis. A Cover, Test Set MX-9490/APM-305A shall be provided with the instrument case for transportability (See 4.20). Any requirement for projecting feet for stacking purposes shall not apply.

3.10.1 ORIENTATION. The instrument case shall permit installation of the equipment chassis within the case in only one position (See 4.20).

3.10.2 CHASSIS FASTENERS. The equipment shall be contained within the instrument case by means of captive screws that permit secure fastening and ease in removal (See 4.20). Screws which are used to hold internal parts (switches, knobs, etc) need not be a similar color as the chassis panel.

3.10.3 HANDLES. The instrument case shall have two handles, one at each end, optimally located to facilitate portability of the equipment (See 4.20). Handles shall be in accordance with MIL-T-21200.

3.10.4 COVER. The instrument case cover, Cover, Test Set MX-9490/APM-305A shall protect the front panel controls and indicators and provide storage of the equipment accessories (See 4.20). A schematic diagram and brief operating instructions are not required to be affixed to the inside of the cover or to the chassis.



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3.11 WEIGHT The weight of the equipment, including the instrument case cover and accessories shall not exceed 70 pounds (See 4.10).

3.12 VOLUME. The volume of the equipment, including the instrument case with cover, shall not exceed 2.7 cubic feet. The form factor shall be such that the front panel dimensions shall not exceed 15 inches by 19 inches (See 4.11).

3.13 GOVERNMENT FURNISHED PROPERTY. Government furnished property will be furnished to the contractor as specified in the contract.

3.14 COMPATIBILITY. (See 4.12).

3.14.1 IFF SYSTEM COMPATIBILITY. The equipment shall be compatible with the IFF Mark X(SIF)/Air Traffic Control Radar Beacon System (Mark X SIF/ATCRBS), in accordance with U.S. National Standards 1010.51A for Common System Component Characteristics and Specification DOD-AIMS 65-1000 (See 4.12).

3.14.2 EQUIPMENT COMPATIBILITY. The equipment, when used in conjunction with the AN/APM-239 Transponder Bench Test Set and the AN/USM-281 Oscilloscope, or equivalent, shall be operationally compatible with the AN/APX-72 and AN/APX-100 Transponder Sets (See 4.12).

3.14.3 CHASSIS GROUND. A separate front panel terminal, shall be provided to permit grounding the equipment chassis to the chassis of the accompanying oscilloscope (See 4.12).

3.15 RELIABILITY. The equipment shall have a mean-time-between-failure (MTBF) of 900 hours when subjected to the environmental conditions of sec. 4.5.5.

3.16 ELECTROMAGNETIC INTERFERENCE EMISSION AND SUSCEPTIBILITY. The equipment shall meet the following requirements of MIL-STD-461A, Notice 4 (See 4.13):

| | | | |
|---|---|---|---|
| CE02 | CS02 | RE02 | RS02 |
| CE04 | CS06 | RE02.1 | RS03* |
| | | | RS03.1* |

*The limits for nonsheltered equipment shall apply.

3.17 WARM-UP TIME. The equipment warm-up time shall not exceed ten minutes when operating at room ambient and at extreme operating temperature conditions (See 4.14).

3.18 SERVICE CONDITIONS. The equipment shall conform to the requirements of this specification for the following service conditions (See 4.15).

3.18.1 TEMPERATURE. (See 4.15.1).



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3.18.1.1 HIGH TEMPERATURE. The equipment shall meet the specified performance requirements of Table IV after storage at 85°C and when operated at 55°C. The time duration of high temperature storage at 85oC shall be 48 hours (See 4.15.1.1).

3.18.1.2 LOW TEMPERATURE. The equipment shall meet the specified performance requirements of Table IV after at -62°C and when operated at 0°C. the time duration of low temperature storage at -62oC. shall be 4 hours (See 4.15.1.2).

3.18.2 ALTITUDE. The equipment shall meet the specified performance requirements of Table IV when operating up to 10,000 feet above sea level and withstand, in an non-operation condition, exposure up to 50,000 feet above sea level (See 4.15.2).

3.18.3 HUMIDITY. The equipment shall withstand the effect of humidities up to 100 percent, including conditions wherein condensation takes place in and on the equipment. The equipment shall withstand the above conditions during operating and non-operating conditions (See 4.15.3).

3.18.4 VIBRATION. The equipment in a non-operating condition shall be capable of withstanding vibration encountered during service conditions (See 4.15.4).

3.18.5 SHOCK. (See 4.15.5).

3.18.5.1 BENCH HANDLING. The equipment shall be capable of withstanding shock encountered in servicing (See 4.15.5.1).

3.18.5.2 BASIC DESIGN. The equipment shall be capable of withstanding shock induced by routine handling (See 4.15.5.2).

3.18.6 FUNGUS. The equipment shall be resistant to fungus growth (See 4.15.6).

3.18.7 SALT FOG. The equipment shall be capable of withstanding exposure in normal operating configuration, to salt fog and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish, or clogging or binding of moving parts (See 4.15.7).

3.19 ELECTRICAL REQUIREMENTS, TEST SET, TRANSPONDER TS-3395/APM-305A. Unless otherwise specified, the equipment shall be capable of meeting all the following requirements (See 4.16):

3.19.1 PULSE REPETITION FREQUENCY(PRF). (See 4.16.1).

3.19.1.1 PRF SYNCHRONIZATION. One challenge signal (3.19.4.3) shall be generated in synchronism with each PRF cycle (See 4.16.1.1).

3.19.1.1.1 FIXED FREQUENCY MODE. (See 4.16.1.1.1).



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3.19.1.1.1.1 BASIC PRF. For the fixed frequency mode of operation, the equipment shall operate at a basic PRF of from 10 to 10,000 pps as selected by front panel controls (See 4.16.1.1.1.1).

3.19.1.1.1.2 REDUCED PRF. When selected by means of a front panel switch, the internal PRF shall be reduced to one half of basic PRF to cover the range of 5 to 5000 pps (See 4.16.1.1.1.2).

3.19.1.1.2 SWEEP FREQUENCY MODE. For the sweep frequency modes of operation, the equipment shall operate at the following repetition rates (See 4.16.1.1.2):

| SWEEP BANDWIDTH | PRF |
|---|---|
| Narrow Band(±5 MHz) | 800 Hz ±10% |
| Wide Band(±20 MHz) | 250 Hz ±10% |

3.19.2 "0"(ZERO) TRIGGER. The equipment shall generate an internal synchronization trigger with the leading edge used as a zero time reference (See 4.16.2).

3.19.3 TRIGGER OUTPUT. (See 4.16.3).

3.19.3.1 SCOPE TRIGGERS. For all modes of operation, an output trigger shall be provided to trigger the Oscilloscope AN/USM-281, or equivalent. The output trigger pulse shall have the following characteristics (See 4.16.3.1):

| | | |
|---|---|---|
| a. | Amplitude | 5.0 ±1.0 volts |
| b. | Impedance (Of Load) | 100 ohms |
| c. | Rise Time | Less than 0.1 USEC |
| d. | Decay Time | Less than 0.5 USEC |
| e. | Width | 1.0 to 5.0 USEC |

The trigger shall be one of the three types specified in 3.19.3.1.1 thru 3.19.3.1.3.

3.19.3.1.1 EXTERNAL VARIABLE DELAY TRIGGER. For the fixed frequency mode of operation, an output trigger shall be provided which is continuously variable by front panel controls from 0.5 USEC to 4000 USEC following the "0" trigger (See 4.16.3.1.1).

3.19.3.1.1.1 JITTER. The jitter of the external variable delay trigger shall be less than ±0.01 USEC ±0.01% of the delay time with respect to the "0" trigger (See 4.16.3.1.1.1).

3.19.3.1.2 FIXED TRIGGER DUAL 3/A. For Dual 3/A challenge video (3.19.4.3.8), the output trigger shall occur within ±0.5 USEC of the "0" trigger (See 4.16.3.1.2).



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3.19.3.1.3 SWEEP SYNC TRIGGER. For the sweep frequency modes of operation, a sweep sync output trigger shall be within 1 USEC of the start of each sweep (See 4.16.3.1.3).

3.19.3.2 AUXILIARY TRIGGER. An auxiliary trigger output controlled by front panel "ON-OFF" switch shall be provided with the following characteristics (See 4.16.3.2):

| | | |
|---|---|---|
| a. | Amplitude | 20.0 ±3 volts with 90 ±10% ohms load. |
| b. | Pulse Width | 1.0 ±0.5 USEC |
| c. | Rise Time | Less than 0.1 USEC |
| d. | Decay Time | Less than 0.2 USEC |

3.19.3.2.1 AUXILIARY TRIGGER IN SWEEP FREQUENCY MODES. The auxiliary trigger shall occur within 0.5 USEC of the external variable delay trigger in the sweep frequency modes (See 4.16.3.2.1).

3.19.3.2.2 AUXILIARY TRIGGER IN FIXED FREQUENCY MODE. The auxiliary trigger shall occur 4.0 ±1.0 USEC after the "Q" trigger in the fixed frequency modes (See 4.16.3.2.2).

3.19.4 CHALLENGE VIDEO. The equipment shall generate encoded IFF challenge pulse trains and video signals for internal modulation (3.19.10.8) (See 4.16.4).

3.19.4.1 FIXED FREQUENCY. For fixed frequency operation, challenge video signals shall be generated in synchronization with the "0" trigger (See 4.16.4.1).

3.19.4.2 SWEEP FREQUENCY. For sweep frequency operation, challenge video signals shall be generated in synchronization with the variable delay trigger (See 4.16.4.2).

3.19.4.2.1 CHALLENGE VIDEO DELAY CONTROL. The front panel controls specified in paragraph 3.19.3.1.1 shall also delay the triggering of the challenge video pulses from 0.5 USEC to 4000 USEC following the sweep sync trigger (3.19.3.1.3) (See 4.16.4.2.1)

3.19.4.3 CHALLENGE VIDEO SELECT. A front panel control shall select challenge video signals specified in 3.19.4.3.1 through 3.19.4.3.9 (See 4.16.4.3).

3.19.4.3.1 CHALLENGE - MODE 1. Two pulses having a nominal spacing of 3.0 ±0.05 USEC between leading edges shall be generated (See 4.16.4.3.1).

3.19.4.3.2 CHALLENGE - MODE 2. Two pulses having a nominal spacing of 5.0 ±0.05 USEC between leading edges shall be generated (See 4.16.4.3.2).



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3.19.4.3.3 CHALLENGE - MODE 3/A. Two pulses having a nominal spacing of 8.0 ±0.05 USEC between leading edges shall be generated (See 4.16.4.3.3).

3.19.4.3.4 CHALLENGE - MODE C. Two pulses having a nominal spacing of 21.0 ±0.05 USEC between leading edges shall be generated (See 4.16.4.3.4).

3.19.4.3.5 CHALLENGE - TEST MODE. Two pulses having a nominal spacing of 6.50 ±0.05 USEC between leading edges shall be generated (See 4.16.4.3.5).

3.19.4.3.6 CHALLENGE - MODE 4 (WORD A). A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 13, 15, 17, 20, 24, 26, 30, 33, 35, 38, 41, 43, 45, 48, 50, 52, 54, 58, 61, 63, 66, 68, and 71 USEC with respect to the leading edge of the first pulse (See 4.16.4.3.6).

3.19.4.3.6.1 CHALLENGE - MODE 4 SPACING ACCURACY. No pulse of the Mode 4 (Word A) pulse train shall vary more than ±0.07 USEC from its nominal value (See 4.16.4.3.6.1.

3.19.4.3.7 CHALLENGE - MODE 4 (WORD B). A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 12, 15, 17, 19, 22, 25, 28, 30, 32, 35, 37, 40, 44, 48, 51, 53, 55, 57, 60, 62, 64, 66, and 70 USEC with respect to the leading edge of the first pulse (See 4.16.4.3.7).

3.19.4.3.7.1 CHALLENGE - MODE 4 SPACING ACCURACY. No pulse of the Mode 4 (Word B) pulse train shall vary more than ±0.07 USEC from its nominal value (See 4.16.4.3.7.1).

3.19.4.3.8 CHALLENGE - DUAL 3/A. Two sets of challenge pulse pairs as specified in 3.19.4.3.3 shall be generated (See 4.16.4.3.8).

3.19.4.3.8.1 CHALLENGE - DUAL 3/A VARIABLE SPACING. The position of the second challenge pair with respect to the first challenge pair shall be variable and controlled by the variable delay trigger (3.19.3.1.1) from 32 USEC to 4000 USEC (See 4.16.4.3.8.1).

3.19.4.3.9 CONTINUOUS WAVES(CW). Continuous wave RF output (3.19.10.8) shall be generated in both modulator channels (See 4.16.4.3.9).

3.19.4.4 VARIABLE CHALLENGE PULSE SPACING. The equipment shall be capable of varying the spacing of the challenge pulses specified in 3.19.4.3.1 through 3.19.4.3.5 and each of the three Mode 4 challenge pulses at the nominal positions of 2,4, and 6 USEC as specified in 3.19.4.3.6 and 3.19.4.3.7 (See 4.16.4.4).

3.19.4.4.1 SPACING SELECT. A front panel switch shall provide selection of the following spacings: 0, +0.2 USEC and +0.90 USEC. The tolerance of all pulse spacings shall be ±0.05 USEC (See 4.16.4.4.1).

3.19.4.4.2 CONTINUOUS SPACING. The equipment shall provide for continuously varying spacing of the movable challenge pulses from -1.0 USEC to +1.0 USEC about the zero position (See 4.16.4.4.2).



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3.19.4.5 CHALLENGE VIDEO PULSE WIDTH. The equipment shall provide a control for varying the width of the challenge pulses specified in 3.19.4.3.1 through 3.19.4.3.8 (See 4.16.4.5).

3.19.4.5.1 WIDTH SELECT. A front panel switch shall be provided for the selection of widths of 0.25 USEC, 0.5 USEC, 0.8 USEC and 1.70 USEC. The tolerance of each pulse width shall be ±0.05 USEC (See 4.16.4.5.1).

3.19.4.5.2 CONTINUOUSLY VARIABLE WIDTH. The equipment shall be provided with a control for continuously varying the width from 0.25 USEC to 1.7 USEC (See 4.16.4.5.2).

3.19.4.6 ISLS CONTROL PULSE (P2). The equipment shall provide a control for combining an ISLS control pulse with the challenge video signal (3.19.4) (See 4.16.4.6).

3.19.4.6.1 ISLS - MODES 1, 2, 3/A, TEST, C and DUAL 3/A. (See 4.16.4.6.1).

3.19.4.6.1.1 NOMINAL SPACING. The ISLS pulse shall be spaced 2.0 ±.05 USEC after the first challenge pulse (P1) (See 4.16.4.6.1.1).

3.19.4.6.1.2 PULSE WIDTH. The pulse width of the ISLS pulse shall be 0.8 ±0.05 USEC (See 4.16.4.6.1.2).

3.19.4.6.2 ISLS - MODE 4 (WORDS A AND B). (See 4.16.4.6.2)

3.19.4.6.2.1 NOMINAL SPACING. The ISLS pulse shall be spaced 8.0 ±0.07 USEC after the first challenge pulse (sync pulse) (See 4.16.4.6.2.1).

3.19.4.6.2.2 PULSE WIDTH. The pulse width of the ISLS pulse shall be 0.5 ±0.05 USEC (See 4.16.4.6.2.2).

3.19.4.6.3 ISLS - DUAL 3/A MODE. For ISLS operation, the ISLS control pulse (P2) shall be added to only the first of the two challenge pulse pairs (See 4.16.4.6.3).

3.19.4.6.4 VARIABLE ISLS PULSE SPACING. The equipment shall provide a control for varying the spacing of the ISLS pulse (P2) with respect to the first challenge pulse (P1) for Modes 1, 2, 3/A, Test, C and Dual 3/A (See 4.16.4.6.4).

3.19.4.6.4.1 SPACING SELECT. A front panel switch shall be capable of providing spacings of 0, ±0.15 USEC and ±0.6 USEC from the nominal position. The tolerance of each pulse spacing shall be ±0.05 USEC (See 4.16.4.6.4.1).

3.19.4.6.4.2 CONTINUOUS SPACING. The equipment shall be provided with a control for continuously varying the spacing of the ISLS pulse ±1.0 USEC of its nominal position (See 4.16.4.6.4.2).



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3.19.5 SUPPRESSION PULSE OUTPUT. The equipment shall provide, as a front panel ON-OFF-switch-controlled output, a suppression pulse having the following characteristics (See 4.16.5):

| | | |
|---|---|---|
| a. | Amplitude | 20 ±2 volts positive with a load of 2200 ohms ±10% resistance in parallel with 50 picofarad or less capacitance. |
| b. | Rise Time | The leading edge of the pulse shall have a slope of at least 20 V/USEC. |
| c. | Decay Time | The trailing edge shall have a slope of at least 20 V/USEC. |
| d. | Duration | 30 ±3 USEC. |
| e. | Timing | The 50% point of the leading edge of the pulse shall occur within 1.0 USEC of the 50% point of the leading edge of the zero trigger. |
| f. | Duty Cycle | The duty cycle of the suppression pulse need not exceed 20%; however, provisions shall be made to preclude damage to the equipment regardless of the PRF selected. |

3.19.6 MODE 4. (See 4.16.6).

3.19.6.1 MODE 4 ENABLE TRIGGER. The equipment shall accept a transponder Mode 4 enable trigger having the following characteristics (See 4.16.6.1):

| | | |
|---|---|---|
| a. | Amplitude | 1.5 to 5.0 volts positive. |
| b. | Width | 0.5 USEC to 3.0 USEC. |
| c. | Rise Time | 0.1 USEC or less. |
| d. | Decay Time | 1.0 USEC or less. |
| e. | Impedance (of Load) | 90 ohms ±10% |

3.19.6.2 MODE 4 REPLY OUTPUT. For each enable trigger input, a Mode 4 video reply output shall be generated (except when inhibited in accordance with 3.19.6.4) with the following characteristics (See 4.16.6.1):



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| | | |
|---|---|---|
| a. | Amplitude | 5 ±0.5 volts, positive, across 90 ohms ±10% |
| b. | Width | 0.5 ±0.1 USEC |
| c. | Spacing | 3 pulses spaced by 1.8 ±0.1 USEC |
| d. | Timing | 200 USEC ±5 USEC after the enable trigger |

3.19.6.3 MODE 4 VIDEO INPUT. The equipment shall accept from the transponder under test, Mode 4 challenge video having the following characteristics (See 4.16.6.2):

| | | |
|---|---|---|
| a. | Amplitude | 1.5 to 5 volts, positive |
| b. | Width | 0.4 to 0.6 USEC |
| c. | Number of Pulses | Up to 37 pulses spaced 2 USEC apart |
| d. | Impedance (of Load) | 90 ohm nominal |
| e. | Pulse Train Droop | 15 percent maximum |

3.19.6.4 MODE 4 DISPARITY PULSE CHARACTERISTICS. When disparity is selected (3.19.6.4.1) or an ISLS pulse is present in the Mode 4 video input, the Mode 4 reply output shall be inhibited and the equipment shall generate a disparity pulse with the following characteristics (See 4.16.6.2):

| | | |
|---|---|---|
| a. | Amplitude | 5 ±0.5 volts, positive |
| b. | Pulse Width | 0.5 ±0.1 USEC |
| c. | Timing | 65 ±1.0 USEC after the enable trigger (fourth sync pulse of a Mode 4 interrogation) for ISLS disparity or coincident with the pulse at 66 USEC for Mode 4 (Words A and B) disparity |
| d. | Impedance (of Load) | 90 ohms ±10% |

3.19.6.4.1 INHIBIT CONTROL SWITCH. A three position switch shall be provided on the front panel to perform the following (See 4.16.6.2):

a. OFF. All ISLS and disparity functions inoperative.

b. ISLS-ON The ISLS pulse (3.19.4.6.2) shall be provided as specified herein.

c. Disparity(MOM). This shall be a spring loaded position such that when engaged, the pulse in the 66 USEC position of the Mode 4 maintenance word selected shall be deleted.



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3.19.6.5 MODE 4 INTERFACE CONNECTOR. The front panel mounted Mode 4 interface connector shall be Type ONO-89559-3 or equivalent (See 4.16.6.3).

3.19.7 MARKERS. (See 4.16.7).

3.19.7.1 FIXED FREQUENCY.(See 4.16.7.1)

3.19.7.1.1 SELECTIVE IDENTIFICATION FEATURE(SIF) REPLY MARKERS. Markers shall be provided at a front panel output connector with a spacing of 20.3 ±0.02 USEC, 24.65 ±0.02 USEC and 49.3 ±0.02 USEC from the initial marker (See 4.16.7.1.1).

3.19.7.1.1.1 MARKER PHASING. A front panel control shall permit movement of all markers simultaneously (See 4.16.7.1.1.1).

3.19.7.1.1.2 INITIAL MARKER TIMING. The initial marker shall be capable of being adjusted within the range of 2.0 USEC to 4.0 USEC after the last challenge pulse by the marker phasing control (See 4.16.7.1.1.2).

3.19.7.1.1.3 CHARACTERISTICS. The SIF markers shall have the following characteristics (See 4.16.7.1.1.3):

| | | |
|---|---|---|
| a. | Amplitude | 0.5 volt, positive, or greater. |
| b. | Impedance (of Load) | 75 ohms |
| c. | Width | 0.15 ±0.05 USEC |

3.19.7.1.2 TIMING MARKERS. Markers, synchronized with the challenge pulse trains(3.19.4), shall be provided at a separate front panel output connector (See 4.16.7.1.2).

3.19.7.1.2.1 MARKER SPACING. The markers shall be spaced in intervals of 0.1, 1.0 and 10.0 USEC with accuracy of ±0.02% (See 4.16.7.1.2.1).

3.19.7.1.2.2 CHARACTERISTICS. The timing markers shall have the following characteristics (See 4.16.7.1.2.2):

| | | |
|---|---|---|
| a. | Amplitude | 0.5 volt, positive, or greater. |
| b. | Impedance (of Load) | 75 ohms. |
| c. | Width | No more than 35% of time intervals between markers. |

3.19.7.2 SWEEP FREQUENCY. Frequency markers shall be provided at a front panel marker output connector for use with the equipment RF sweep frequency circuits (See 4.16.7.2).



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3.19.7.2.1 NARROW SWEEP BANDWIDTH. A marker shall be provided at center frequency, with additional markers at -5 MHz, -3 MHz, -1 MHz, +1 MHz, +3 MHz and +5 MHz about the center frequency (See 4.16.7.2.1).

3.19.7.2.2 WIDE SWEEP BANDWIDTH. Markers shall be provided the same as narrow sweep bandwidth with the addition of markers at -20 MHz and +20 MHz about the center frequency (See 4.16.7.2.2).

3.19.7.2.3 MARKER ACCURACY. Marker accuracy shall be ±0.1 MHz for all markers except for -20 MHz and +20 MHz markers, for which it shall be ±0.3 MHz (See 4.16.7.2.3).

3.19.7.2.4 CHARACTERISTICS. The frequency markers shall have the following characteristics (See 4.16.7.2.4):

| | | |
|---|---|---|
| a. | Amplitude | 5.0 volts ±1.0 volt positive. |
| b. | Width | 2.0 USEC ±0.5 USEC. |
| c. | Impedance (of Load) | 10K ohm. |

3.19.8 RF SIGNAL GENERATION. The equipment shall provide the following fixed frequency and sweep frequency RF output signals (See 4.16.8).

3.19.8.1 FIXED FREQUENCY. The equipment shall provide a crystal controlled 1030 MHz output signal with an accuracy of ±0.01 percent (See 4.16.8.1).

3.19.8.2 SWEEP FREQUENCY. The equipment shall contain RF frequency sweep capabilities (See 4.16.8.2).

3.19.8.2.1 SWEEP BANDWIDTH. (See 4.16.8.2.1).

3.19.8.2.1.1 NARROW BAND. The equipment shall sweep from 1025 MHz to 1035 MHz (See 4.16.8.2.1.1).

3.19.8.2.1.2 WIDE BAND. The equipment shall sweep from 1010 MHz to 1050 MHz (See 4.16.8.2.1.2).

3.19.8.2.1.3 SWEEP BLANKING. For RF sweep operation, modulation of the equipment RF output shall only occur during the time the RF sweep changes frequency (See 4.16.8.2.1.3).

3.19.9 LOW LEVEL RF INPUT. The equipment shall be provided with a low level RF input channel whose input connector is of a different type from the high level input (3.19.10.1) to minimize the possibility of connecting a high level signal to this input (See 4.16.9).



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3.19.10 HIGH LEVEL RF INPUT/OUTPUT. The equipment shall provide two RF input/output channels, specified as MAIN and AUXILLIARY (See 4.16.10).

3.19.10.1 RF INPUT/OUTPUT CHANNEL SWITCH. The equipment shall be provided with a front panel switch to select either of the two RF channels for demodulation (See 4.20).

3.19.10.2 CONNECTORS. Two front panel Type "N" female connectors shall be provided for the equipment RF input/output channels (See 4.20).

3.19.10.3 INPUT/OUTPUT VSWR. The VSWR of each channel shall not exceed 1.35:1 with a load impedance of 50 ohms (See 4.16.10.1).

3.19.10.4 ATTENUATION CONTROLS. Two separate front panel attenuators shall be provided for continuously varying each of the equipment RF output levels between 10 and 90 dB below 1 milliwatt when matched to a 50 ohm impedance load (See 4.16.10.2).

3.19.10.4.1 ATTENUATOR. Each attenuator shall be calibrated in dB below 1 milliwatt under matched load conditions (See 4.16.10.2.1).

3.19.10.4.2 ATTENUATOR SCALES. Each attenuator scale shall contain 80 divisions at one dB intervals and be labeled at each tenth division, i.e., -10, -20, -30, -40, -50, -60, -70, -80, and -90 (See 4.20).

3.19.10.4.3 DIFFERENTIAL ATTENUATION. The differential attenuation between any two consecutive 10 dB divisions shall be 10.0 ±0.5 dB (See 4.16.10.2.2).

3.19.10.5 AUXILIARY ATTENUATOR. The attenuator for the AUXILIARY channel shall have a separate ISLS scale and an adjustable indicator to provide a direct indication of the amplitude of the ISLS pulse (P2) relative to the amplitude of the challenge pulses (See 4.16.10.3).

3.19.10.5.1 ISLS SCALE. The scale shall be labeled ISLS with markings of one dB divisions from 0 to -9 dB and shall correlate with basic auxiliary markings (3.19.10.4.2) (See 4.20).

3.19.10.5.2 ADJUSTABLE INDICATOR. An adjustable indicator shall be provided to permit setting the indicator index to 0 dB when the basic scale is set to -10 dBm to -12 dBm (See 4.16.10.3).

3.19.10.6 RF OUTPUT LEVEL ACCURACY. (See 4.16.10.4).

3.19.10.6.1 FIXED FREQUENCY. The equipment output level for fixed frequency operation and nominal pulse width shall be accurate within ±1.0 dB of the attenuator setting (See 4.16.10.4.1).



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3.19.10.6.2 SWEEP FREQUENCY. The equipment output level for sweep frequency operation shall be flat within 1.0 dB peak-to-peak over the sweep bandwidth (See 4.16.10.4.2).

3.19.10.6.2.1 SWEEP OFFSET. For the same attenuator setting as for fixed frequency operation (3.19.10.6.1) the output level of the RF sweep at 1030 MHz center frequency shall be within ±0.5 dB of the fixed frequency level (See 4.16.10.4.2.1).

3.19.10.7 ISLS RF OUTPUT. The equipment RF output for ISLS operation shall be provided from the MAIN input/output channel (See 4.16.10.5).

3.19.10.7.1 MAIN LEVEL. The MAIN attenuator shall control the level of the combined challenge video signal (P1, P2 and P3) from -10 dBm to -90 dBm (See 4.16.10.5.1).

3.19.10.7.2 AUXILIARY LEVEL. The AUXILIARY attenuator shall control the level of the ISLS pulse (P2) within the combined video signal from an amplitude of 0 dB (equal to the amplitude of the P1 and P3 pulses) to at least 70 dB below the level of the P1 and P3 pulses (See 4.16.10.5.1).

3.19.10.7.3 ISLS LEVEL SET ACCURACY. With the AUXILIARY attenuator scale indicating an ISLS pulse amplitude equal to the P1 and P3 pulse amplitude, the ISLS level set accuracy shall be +2 dB, -0 dB relative to the amplitude of the P1 and P3 pulses (See 4.16.10.5.1).

3.19.10.8 MODULATION. The equipment's two RF outputs shall be capable of being separately pulse modulated with challenge pulse trains as specified in 3.19.4.3.1 through 3.19.4.3.9 (See 4.16.10.6).

3.19.10.8.1 MODULATION SELECTION. The equipment modulation shall be selected by means of a front panel control (See 4.16.10.6).

3.19.10.8.2 CW LEAKAGE. When pulse modulation is selected, the equipment's CW leakage from either RF input/output channel shall be no greater than -90 dBm (See 4.16.10.6.1).

3.19.10.8.3 MODULATOR ON-OFF RATIO. The equipment modulators shall have on-to-off ratio of 60 dB or greater (See 4.16.10.6.2).

3.19.10.8.4 AUXILIARY CHANNEL DELAY RANGE. A front panel control shall be provided to delay the output of the AUXILIARY channel from 0.05 USEC to 0.4 USEC with respect to the MAIN channel (See 4.16.10.6.3).

3.19.10.8.4.1 AUXILIARY CHANNEL DELAY CONTROL DETENT. The control shall have a detent at 0.2 ±0.05 USEC (See 4.16.10.6.3.1).



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3.19.10.8.5 MODULATION FIDELITY. The equipment's RF output signals shall have the following characteristics when modulated by the challenge pulse trains (See 4.16.10.6.4).

| | | |
|---|---|---|
| a. | Rise Time | 0.05 to 0.1 USEC. |
| b. | Decay Time | 0.05 to 0.2 USEC. |
| c. | Width | As specified in 3.19.4.5 and 3.19.4.6. |

3.19.11 MEASUREMENT CAPABILITY. The equipment when used with the AN/USM-281 Oscilloscope or equivalent shall be capable of measuring pulse fidelity, RF power, RF frequency and PRF characteristics of IFF transponder reply signals (See 4.16.11).

3.19.11.1 DEMODULATOR SENSITIVITY. (See 4.16.11.1).

3.19.11.1.1 HIGH LEVEL. The equipment shall demodulate an RF input of +18 dBw to 33 dBw and shall produce a front panel demodulated video output pulse of not less 1 volt amplitude into a 75 ohm impedance (See 4.16.11.1.1).

3.19.11.1.2 LOW LEVEL. The equipment shall demodulate an RF input of -12 dBm to +3 dBm and shall produce a front panel demodulated video output pulse of not less than 1 volt amplitude into a 75 ohm impedance (See 4.16.11.1.2).

3.19.11.1.2.1 EXTENDED LOW LEVEL RANGE. The low power input capabilities shall be extended to cover the range of +3 dBm to +48 dBm using the attenuators specified in 3.20.6 (See 4.16.11.3).

3.19.11.2 PULSE FIDELITY. The distortion in the demodulated video output's pulse rise time, fall time and width shall not deviate from a corresponding output of a standard biased detector by more than ±0.02 USEC or ±5%, whichever tolerance is greater, over the frequency range of 1085 MHz to 1095 MHz (See 4.16.11.2).

3.19.11.3 PULSE POWER MEASUREMENT CAPABILITY. The equipment shall have the following pulse power measurement capability with the readout indicated on the equipment meter specified in 3.19.11.6 for the high level inputs (3.19.11.1.1) and with a conversion chart (3.20.8) and meter indicator for the low level inputs (3.19.11.1.2 and 3.19.11.2.1) (See 4.16.11.3).



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| | TOLERANCE | MINIMUM | MAXIMUM |
|---|---|---|---|
| Power Range | --- | -12 dBm | 63 dBm |
| Frequency Range | --- | 1085 MHz | 1095 MHz |
| Accuracy High Level | ±0.5 dB | 25.5 dBw | 29.0 dBw |
| Accuracy High Level | ±1.0 dB | 18.0 dBw | 25.49 dBw |
| Accuracy High Level | ±1.0 dB | 29.1 dBw | 33.0 dBw |
| Accuracy Low Level | ±1.0 dB | -12 dBm | 3 dBm |
| Accuracy Extended Low Level | ±1.0 dB* | 3.1 dBm | 47.9 dBm |

*Plus accuracy of the accessory attenuators.

3.19.11.3.1 POWER BIAS CONTROL. For the RF input power levels to be measured, a front panel control shall reduce the amplitude of the demodulated video output pulse to a 1.0 volt level (See 4.16.11.3.1).

3.19.11.4 FREQUENCY MEASUREMENT. The equipment shall have the following frequency measurement capability using AN/USM-281 or equal for a readout indicator (See 4.16.11.4):

| | TOLERANCE | MINIMUM | MAXIMUM |
|---|---|---|---|
| Frequency Range | --- | 1070 MHz | 1110 MHz |
| Accuracy | ±0.6 MHz | 1070.0 MHz | 1086.9 MHz |
| Accuracy | ±0.2 MHz | 1087.0 MHz | 1093.0 MHz |
| Accuracy | ±0.6 MHz | 1093.1 MHz | 1110.0 MHz |

3.19.11.5 PRF MEASUREMENT. The equipment shall be capable of measuring the repetition frequency of the internal operating PRF and the demodulator output to an accuracy of ±5% or ±5 degrees, whichever tolerance is greater, from 1/10th full scale to full scale (See 4.16.11.5).

3.19.11.6 METER. The equipment shall contain a multi-scale meter with scales marked in Hz and dBw to indicate challenge/reply PRF's and power levels (See 4.16.11.3 and 4.16.11.5).

3.19.11.7 SELF-TEST. (See 4.16.11.6).

3.19.11.7.1 SIGNAL ROUTING. The internally generated RF signals shall be routed from the RF input/output channel connectors (MAIN, AUXILIARY) to the low power input connector through an RF jumper cable specified in 3.20.3 (See 4.16.11.6.1).

3.19.11.7.1.1 PULSE FIDELITY MEASUREMENT. The pulse rise time, fall time and width characteristics of 3.19.10.8.5 shall not be degraded by more than 0.02 USEC or ±5%, whichever tolerance is greater, when the equipment's RF outputs are demodulated during self-test operation (See 4.16.11.6.1.1).



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3.19.11.7.1.2 POWER MEASUREMENT ACCURACY. The measurement accuracy of the equipment peak power output at the -10 dBm level shall be ±0.5 dB (See 4.16.11.6.1.2).

3.19.11.7.2 BUILT-IN-TEST(BIT) FOR FREQUENCY MEASUREMENT. The equipment shall have a built-in capability to measure the accuracy of its frequency measurement circuitry to a tolerance of ±0.3 MHz (See 4.16.11.6.2).

3.19.12 POWER SUPPLY FAULT INDICATOR. The equipment shall have a front panel indicator which will illuminate when any of the DC working voltages developed by the equipment's power supply deviate more than ±1.0 volts from nominal. The indicator shall not illuminate for variations within ±0.25 volts of nominal. For variations between ±0.25 volts and ±1.0 volts of nominal, the indicator may or may not illuminate (See 4.16.12).

3.19.13 POWER SUPPLY SHORT CIRCUIT PROTECTION. The equipment shall have the capability to prevent power supply damage when any of the DC working voltage lines become shorted (See 4.16.13)..

3.20 ACCESSORIES. (See 4.17).

3.20.1 RF CABLE. Two 8-foot ±0.25 ft. lengths of RF cable, type RG-212/U having a UG-18/U, or equivalent, plug on each end, shall be provided. This cable shall be marked to indicate its attenuation at 1030 MHz and 1090 MHz (See 4.17.1).

3.20.2 VIDEO CABLES. Four 5-foot ±0.25 ft. lengths of RF cables, type RG-62A/U, each having a UG-260F/U plug on each end, shall be provided (See 4.17.1).

3.20.3 RF JUMPER CABLE. One 12-inch ±1 inch RF cable RG-223/U, having suitable connectors to connect the MAIN and/or AUXILIARY Channels to the equipment low power input/output for self-test purposes, shall be provided (See 4.17.1).

3.20.4 PRIMARY POWER CABLE. One 8-foot ±0.25 ft. length of AC power cable with suitable connectors shall be provided (See 4.17.1).

3.20.5 ADAPTERS. Two Type UG-565A/U "C" (Male) to "N" (Female) adapters shall be provided (See 4.20).

3.20.6 FIXED ATTENUATORS. A 15.0 ±0.2 dB attenuator and a 30.0 ±0.3 dB attenuator, each having a connector to mate with the low power input connector (3.19.9) on one end and a connector functionally and electrically equivalent to the low power input connector on the other end shall be provided with the following characteristics (See 4.16.11.3 and 4.17.3).



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| | | |
|---|---|---|
| a. | Frequency Range | 1070 MHz to 1110 MHz. |
| b. | Impedance | 50 ±3 ohms. |
| c. | VSWR (Maximum) | 1.2:1. |
| d. | Power Dissipation Rating | 1.0 watt or greater. |

3.20.7 VIDEO TERMINATIONS. Three feed through-type terminations, each having a connector on one end to mate with a UG-912 and a connector on the other end to mate with a UG-260, shall be provided as follows: (See 4.17.2):

| QUANTITY | DC RESISTANCE |
|---|---|
| 2 | 75 ohms ±5% |
| 1 | 93 ohms ±5% |

3.20.8 CONVERSION CHART . A chart shall be provided to convert from the meter indication to the actual input power level applied to the low power input connector (See 4.20).

3.21 OPERABILITY. The unit(s) shall be operable as specified, after applicable processes and inspections are completed (See 4.18).

3.22 INTERCHANGEABILITY. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the units (See 4.19).

3.23 WORKMANSHIP. The equipment shall be manufactured and assembled in accordance with the following requirements and paragraphs herein (See 4.20):

MIL-STD-454

| | |
|---|---|
| Cable, Coaxial(RF) | Requirement 65 |
| Cable Multiconductor | Requirement 66 |
| Controls | Requirement 28 |
| Fungus-Inert Materials | Requirement 4 |
| Fuses and Fuse Holders | Requirement 39 |
| Electrical Overload Protection | Requirement 8 |
| Soldering | Requirement 5 |
| Workmanship | Requirement 9 |

4. QUALITY ASSURANCE PROVISIONS.

4.1 RESPONSIBILITY FOR INSPECTION. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as other-



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wise specified in the contract, the contractor may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF INSPECTION. Inspection shall be classified as follows:

a. First Article Inspection. Does not include packaging (See 4.3).

b. Inspection Covered by Subsidiary Documents. (See 4.4).

c. Quality Conformance Inspection.

(1) Quality conformance inspection of equipment before packaging (See 4.5).

(2) Quality conformance inspection of packaging (See 4.21).

4.3 FIRST ARTICLE. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 FIRST ARTICLE UNITS. The contractor shall furnish first article units of the complete Test Set, Transponder Set AN/APM-305A as specified in the contract.

4.3.2 FIRST ARTICLE INSPECTION. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in/debugging and reliability qualification specified in 4.5, and the inspections specified for Group A, Group B, and Group C (See Tables, I, II, III, respectively). The inspection shall be performed in the following order: 4.4, burn-in/debugging, Group A and Group B for all units, and designated units for specific subgroups within Group C, in accordance with 4.5.4.1, and reliability qualification. The reliability qualification may be conducted concurrent with Group C inspection.

4.3.3 FIRST ARTICLE DATA. The first article test plan and test report(s) shall be as required in the contract.

4.4 INSPECTION COVERED BY SUBSIDIARY DOCUMENTS. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:



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| INSPECTIONS | REQT PARA |
|---|---|
| General Requirements | 3.3 |
| Parts, Materials and Processes | 3.4 |
| Finish, Protective | 3.5 |
| Marking | 3.6 |

4.5 QUALITY CONFORMANCE INSPECTION OF EQUIPMENT BEFORE PACKAGING. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 BURN-IN/DEBUGGING. All units presented to the Government for acceptance shall undergo a 48 hour burn-in. The burn-in shall be conducted under the same conditions as the reliability qualification test (4.5.5). The last half of the burn-in period shall be failure free. If a failure occurs during the last half of the burn-in period, the equipment shall be repaired, the repair shall be verified, and the equipment shall be returned to burn-in. This procedure shall be followed until the 24 hour failure free period is achieved. All failures and maintenance actions shall be recorded by the contractor and the records shall be supplied to the Government as specified in the contract.

4.5.2 GROUP A INSPECTION. Each unit on contract shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table I.

4.5.2.1 ORDER OF INSPECTION WITHIN GROUP A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.3 GROUP B INSPECTION. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.3.1 GROUP B SAMPLING PLANS. The Group B AQL shall be 6.5% and the inspection level shall be S4.

4.5.3.2 ORDER OF INSPECTION WITHIN GROUP B. Group B inspection shall be performed in an order satisfactory to the Government.



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4.5.4 GROUP C INSPECTION. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. No defects shall be allowed.

4.5.4.1 SAMPLING FOR GROUP C INSPECTION. Sample units shall be selected at random for each group of inspections in Table III. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot(s).

4.5.4.1.1 SUB-GROUP I. For this subgroup, two units from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent inspections two (2) units from each fifty (50) production units shall be selected.

TABLE I - GROUP A INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH | AQL MAJOR | AQL MINOR |
|---|---|---|---|---|
| Visual and | 3.23 | 4.20 | 1% | 4% |
| Printed Circuit Board Assemblies | | | 1% | 4% |
| TS-3995/APM-305A | | | 1% | 4% |
| Conntector Adapter | | | 1% | 4% |
| Termination | | | 1% | 4% |
| Conversion Chart | | | 1% | 4% |
| Case, Instrument | | | 1% | 4% |
| Cable Assemblies | | | 1% | 4% |
| An/APM-305A | | | 1% | 4% |
| Electrical | | | | |
| Fixed Frequency Mode | 3.19.1.1.1.1 thru 3.19.1.1.1.2 | 4.16.1.1.1.1 thru 4.16.1.1.1.2 | | |
| Sweep Frequency Mode | 3.19.1.1.2 | 4.16.1.1.2 | | |
| Scope Triggers | 3.19.3.1 thru 3.19.3.1.3 | 4.16.3.1 thru 4.16.3.1.3 | | |
| Challenge Video Select | 3.19.4.3 thru 3.19.4.3.9 | 4.16.4.3 thru 4.16.4.3.9 | | |



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TABLE I - GROUP A INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH | AQL MAJOR | AQL MINOR |
|---|---|---|---|---|
| Variable Challenge Pulse Spacing | 3.19.4.4 thru 3.19.4.4.2 | 4.6.4.4 thru 4.16.4.4.2 | | |
| Challenge Video Pulse Width | 3.19.4.5 thru 3.19.4.5.2 | 4.16.4.5 thru 4.16.4.5.2 | | |
| ISLS Control Pulse | 3.19.4.6 thru 3.19.4.6.4.2 | 4.16.4.6 thru 4.16.4.6.4.2 | 2.5% for the Combined Group | |
| Suppression Pulse Output | 3.19.5 | 4.16.5 | | |
| Mode 4 | 3.19.6.1 thru 3.19.6.5 | 4.16.6.1 thru 4.16.6.3 | | |
| Markers | 3.19.7.1.1 thru 3.19.7.2.4 | 4.16.7.1.1 thru 4.16.7.2.4 | | |
| RF Signal Generation | 3.19.8.1 thru 3.19.8.2.1.2 | 4.16.8.1 thru 4.16.8.2.1.2 | | |
| Attenuation Controls | 3.19.10.4 | 4.16.10.2 | | |
| Auxiliary Attenuator | 3.19.10.5 thru 3.19.10.5.2 | 4.16.10.3 | | |
| Fixed Frequency | 3.19.10.6.1 | 4.16.10.4.1 | | |
| Sweep Frequency | 3.19.10.6.2 | 4.16.10.4.2 | | |
| ISLS RF Output | 3.19.10.7 thru 3.19.10.7.3 | 4.16.10.5.1 | | |
| Modulation | 3.19.10.8 thru 3.19.10.8.1 | 4.16.10.6 | | * |
| Auxiliary Channel Delay Rnage | 3.19.10.8.4 | 4.16.10.6.3 | | |
| Auxiliary Channel Control Detent | 3.19.10.8.4.1 | 4.16.10.6.3.1 | | |
| Modulator Fidelity | 3.19.10.8.5 | 4.16.10.6.4 | | |
| Measurement Capability | 3.19.11 thru 3.19.11.7.2 | 4.16.11 thru 4.16.11.6.2 | | |



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TABLE I - GROUP A INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH | AQL MAJOR | AQL MINOR |
|---|---|---|---|---|
| Cable Assemblies | 3.20.1 thru 3.20.4 | 4.17.1 | 1% | |
| Adapters | 3.20.2 | 4.20 | 1% | 4% |
| Power Switch | 3.8.1 | 4.7.1 | 1% | |
| Power Indicator | 3.8.3 | 4.7.2 | 1% | |
| Operability (Operational Inspection) | 3.21 | 4.18 | 1% | |

*All electrical defects considered major.

TABLE II - GROUP B INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH | AQL |
|---|---|---|---|
| Auxiliary Triggers | 3.19.3.2 thru 3.19.3.2.2 | 4.16.3.2 thru 4.16.3.2.2 | 6.5% for the Combined Group |
| Fixed Frequency | 3.19.3.2.2 | 4.16.4.1 | |
| Sweep Frequency | 3.19.4.2 thru 3.19.4.2.1 | 4.16.4.2 thru 4.16.4.2.1 | |
| Sweep Blanking | 3.19.8.2.1.3 | 4.16.8.2.1.3 | |
| Low Level RF Input | 3.19.9 | 4.16.9 | |
| Differential Attenuation | 3.19.10.4.3 | 4.16.10.2.2 | |
| Sweep Offset | 3.19.10.6.2.1 | 4.16.10.4.2.1 | |
| CW Leakage | 3.19.10.8.2 | 4.16.10.6.1 | |
| Modulator ON-OFF Ratio | 3.19.10.8.3 | 4.16.10.6.2 | |
| Power Supply Fault Indicator | 3.19.12 | 4.16.12 | 6.5% for the Combined Group |
| Power Supply Short Circuit Protection | 3.19.13 | 4.16.13 | |
| Video Terminations | 3.20.7 | 4.17.2 | 6.5% |
| Input Power Source | 3.8 | 4.7 | 6.5% |
| Interchangeability | 3.22 | 4.19 | 6.5% |

4.5.4.1.2 SUB-GROUP II. For this subgroup, one (1) unit from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, on (1) unit shall be selected when one-half of the units on order have been accepted by the Government.



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4.5.4.1.3 SUB-GROUP III. Same as Subgroup II.

4.5.4.1.4 SUB-GROUP IV. Same as Subgroup II.

4.5.4.1.5 SUB-GROUP V. Same as Subgroup II.

4.5.4.1.6 SUB-GROUP VI. For this subgroup, two (2) units shall be selected from the first ten (10) production units. For subsequent Group C inspection, tow (2) units shall be selected when one-half of the units on order have been accepted by the Government.

4.5.4.2 ORDER OF INSPECTION WITHIN GROUP C. Group C inspection shall be performed in the order shown in each subgroup in Table III for Subgroup II thru Subgroup VI. Subgroup I inspection shall be performed in the order satisfactory to the Government.

4.5.4.3 GROUP C FAILURES. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4.4 REINSPECTION OF CONFORMING GROUP C SAMPLE UNITS. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

TABLE III - GROUP C INSPECTION

| INSPECTION | REQUIREMENT PARAGRAPH | TEST PARAGRAPH |
|---|---|---|
| Sub-Group I | | |
| Warm-Up Time | 3.17 | 4.14 |
| PRF Synchronization | 3.19.1.1 | 4.16.1.1 |
| Zero Trigger | 3.19.2 | 4.16.2 |
| Input/Output VSWR | 3.19.10.3 | 4.16.10.1 |
| Front Panel Protection | 3.7 | 4.6 |
| Extender Card | 3.9 | 4.8 |
| Weight | 3.11 | 4.10 |
| Volume | 3.12 | 4.11 |
| Compativility | 3.14 thru 3.14.3 | 4.12 |
| Fixed Attenuators | 3.20.6 | 4.16.11.3 |
| Sub-Group II | | |
| High Temperature | 3.18.1.1 | 4.15.1.1 |
| Low Temperature | 3.18.1.2 | 4.15.1.2 |
| Altitude | 3.18.2 | 4.15.2 |
| Humidity | 3.18.3 | 4.15.3 |
| Sub-Group II | | |
| Vibration | 3.18.4 | 4.15.4 |
| Shock | 3.18.5 | 4.15.5 |



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TABLE III - GROUP C INSPECTION

| | | |
|---|---|---|
| Sub-Group IV<br>Salt Fog | 3.18.7 | 4.15.7 |
| Sub-Group V<br>Fungus | 3.18.6 | 4.15.6 |
| Sub-Group VI<br>EMI | 3.16 | 4.13 |

NOTE: Units selected for Subgroup I inspection may be used for subsequent subgroup testing.

4.5.5 RELIABILITY QUALIFICATION TESTING. The contractor shall conduct a reliability qualification test in accordance with Test Plan V, Test Level B of MIL-STD-781, using a specified MTBF of 900 hours. The test shall be performed on three units (See 3.15).

4.5.5.1 REJECT DECISION. In the event of a reject decision at any time during the test, the paragraph entitled Reject in Reliability Qualification (Demonstration) Phase of MIL-STD-781, shall be followed. The test shall be discontinued, the Government shall be notified, corrective action shall be taken, and a retest shall be done.

4.5.5.2 FAILURE DEFINITION. A failure is defined as any inability of the equipment to perform its required function, including any departure from performance specified herein.

4.5.5.3 PERFORMANCE MEASUREMENTS. The following performance parameters shall be measured at least once daily on each equipment during reliability testing and burn in:

| PARAMETER | REQUIRED PARAGRAPH |
|---|---|
| RF Fixed Frequency Accuracy | 3.19.8.1 |
| Self-Test Power Measurement Accuracy | 3.19.11.7.1.2 |
| Challenge Video Width Select | 3.19.4.5.1 |
| Challenge Video Select - Mode 3/A | 3.19.4.3.3 |
| Challenge Video Select - Mode 4(Word A) | 3.19.4.3.6 |
| ISLS Characteristics - Mode 3/A and 4 (Word A) only | 3.19.4.6.1.1 thru 3.19.4.6.2.2 |
| Frequency Markers | 3.19.7.2.2 |
| Sweep Bandwidth | 3.19.8.2.1 |

4.6 FRONT PANEL PROTECTION. The equipment shall be oriented such that it rests on the front panel handles. Verify that no control, indicator or electrical connector comes in contact with the resting surface (See 3.7).

4.7 INPUT POWER SOURCE: Power shall be applied to the equipment and the tests 4.16.11.6.1.1, 4.16.11.6.1.2, and 4.16.11.6.2 performed at the following combinations of voltages and frequencies (See 3.8):



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| Frequency Hz ( +1%) | Volts VAC( +1%) 103.5 VAC | 115 VAC | 126.5 VAC |
|---|---|---|---|
| 45Hz | X | X | X |
| 60 Hz | | X | |
| 66 Hz | | X | |
| 380 Hz | | X | |
| 420 Hz | X | X | X |

4.7.1 POWER SWITCH Verify that the equipment is energized only when switch is in the "ON" position (See 3.8.1).

4.7.2 POWER INDICATOR. Verify the energizing of power indicator lamp with the application of the AC power source (See 3.8.3).

4.8 EXTENDER CARD. Substitute the extender card for one of the equipment's printed circuit cards. Insert the replaced printed circuit card into the mating connector of the extender card and perform the test of 4.16.11.6.1.2. Verify that all components mounted on the printed circuit card are accessible when the extender card is used. Further verify that keying or other mechanical restraints are provided to prevent improper insertion of the extender card into the replaced printed circuit card position and to prevent improper insertion of the printed card into the extender card's mating connector. Repeat the above for each of the equipment's printed circuit cards (See 3.9).

4.9 STANDARD TEST CONDITION. Unless otherwise specified, inspections shall be performed at prevailing ambient temperature, atmospheric pressure and relative humidity, with the following equipment control settings:

| | |
|---|---|
| Input Power | 115 VAC +10%, 60 Hz +5% |
| Challenge Video | Mode 3/A |
| Challenge Pulse Width | 0.8 USEC |
| Mode of Operation | Fixed Frequency-Dual Channel |
| RF Output | MAIN -10 dBm |
| | AUXILIARY-10 dBm |
| PRF | 1000 Hz |

4.10 WEIGHT. Verify the weight of the equipment (See 3.11).

4.11 VOLUME. Verify the volume of the equipment (See 3.12).

4.12 EQUIPMENT COMPATIBILITY. Connect the equipment front panel ground terminal to the oscilloscope ground terminal. Verify that the equipment chassis is grounded to the oscilloscope chassis. Interconnect the remaining cables (including those from the equipment under test), to the transponder and the AN/APM-239 Test Set as shown in Figure 1. The equipment shall be utilized to measure the following transponder characteristics (See 3.14).

a. Normal Sensitivity.

b. Receiver Bandwidth and Center Frequency.



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c. Transmitter Frequency.

d. Transmitter Power Output.

e. Reply Rate Limiting.

f. ISLS GO/NO-GO Limits.

g. ISLS Decode Characteristics.

h. Reply Frame Spacing and Coding.

i. Auxiliary Trigger Performance.

j. Mode 4 Triggering.

k. Mode 4 Video Out.

l. Mode 4 Replies.

m. Mode 4 Reply Light.

n. Mode 4 Caution Light.

o. Mode 4 Rate Limiting.

p. Mode 4 Sensitivity.

4.13 ELECTROMAGNETIC INTERFERENCE TESTS. The equipment shall be tested for compliance with the EMI requirements for Section 3. The test methods and procedures used shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, Government approved, EMI test plan as specified in the contract (See 3.16).

4.14 WARM-UP. Apply power to the equipment and within ten (10) minutes conduct the test of 4.16.11.6.1.2 (See 3.17).

4.15 SERVICE CONDITIONS. (See 3.18).

4.15.1 TEMPERATURE. (See 3.18.1).

4.15.1.1 HIGH TEMPERATURE. The equipment shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501, Procedure I. Except for the requirement of 3.19.10.6.2, the performance of the equipment during the high temperature test shall be as specified in Table 4 and shall be demonstrated at 55°C with an



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input power source of 126.5 VAC, 420 Hz. During this test, the performance tolerances for requirement 3.19.10.6.2 shall be ±3 dB. The performance of the equipment, before and after high temperature test, shall be as specified in Table 4 and shall be demonstrated under standard Test conditions as specified in 4.9 (See 3.18.1.1).

4.15.1.2 LOW TEMPERATURE. The equipment shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502, Procedure I. Except for the requirement of 3.19.10.6.2, the performance of the equipment during the low temperature test shall be as specified in Table IV and shall be demonstrated at 0°C with an input power source of 103.5 VAC, 45 Hz. During this test, the performance tolerances for requirement 3.19.10.6.2 shall be ±3 dB. The performance of the equipment, before and after the low temperature test, shall be as specified in Table IV and demonstrated under standard test conditions specified in 4.9 (See 3.18.1.2).

4.15.2 ALTITUDE. The equipment shall be subjected to a low pressure test in accordance with MIL-STD-810, Method 500, Procedure I. The performance of the equipment during the low pressure test shall be as specified in Table IV and shall be demonstrated under standard test conditions as specified in 4.9 (See 3.18.2).

4.15.3 HUMIDITY. The equipment shall be subjected to a humidity test in accordance with Method 507, Procedure II of MIL-STD-810. Monitoring tests, for the purpose of determining continued equipment operation, shall be permitted during the last 5-hour period of each humidity cycle in accordance with the instructions specified in Step 6 of Procedure II. The performance during the humidity test shall be determined at Step 8. Except for the requirement of 3.19.10.6.2, the equipment's performance shall be determined in accordance with the test specified in Table IV. The performance tolerances for requirement 3.19.10.6.2 shall be ±3 dB. After the completion of the fifth cycle, the equipment shall be conditioned in accordance with Step 2 and tested as specified in Table IV under standard test conditions. Full performance shall be required (See 3.18.3).

4.15.4 VIBRATION. Prior to initiating the vibration test, the equipment shall be tested in accordance with Table IV. The equipment, while non-operating, shall then be subjected to a simple harmonic motion having an amplitude of 0.015 inches (0.030 inches maximum total excursion), between the limits of 5 to 55 Hz. The rate of frequency change shall be such that a complete cycle (5/55/5) will consume 8 minutes. The test item shall be vibrated for 15 cycles (2 hours) along each of the 3 major axes of the equipment for a total test time of six (6) hours. At the completion of the test, the equipment shall be visually inspected for mechanical failure. The equipment shall then be tested in accordance with the performance tests of Table IV under standard test conditions (See 3.18.4).



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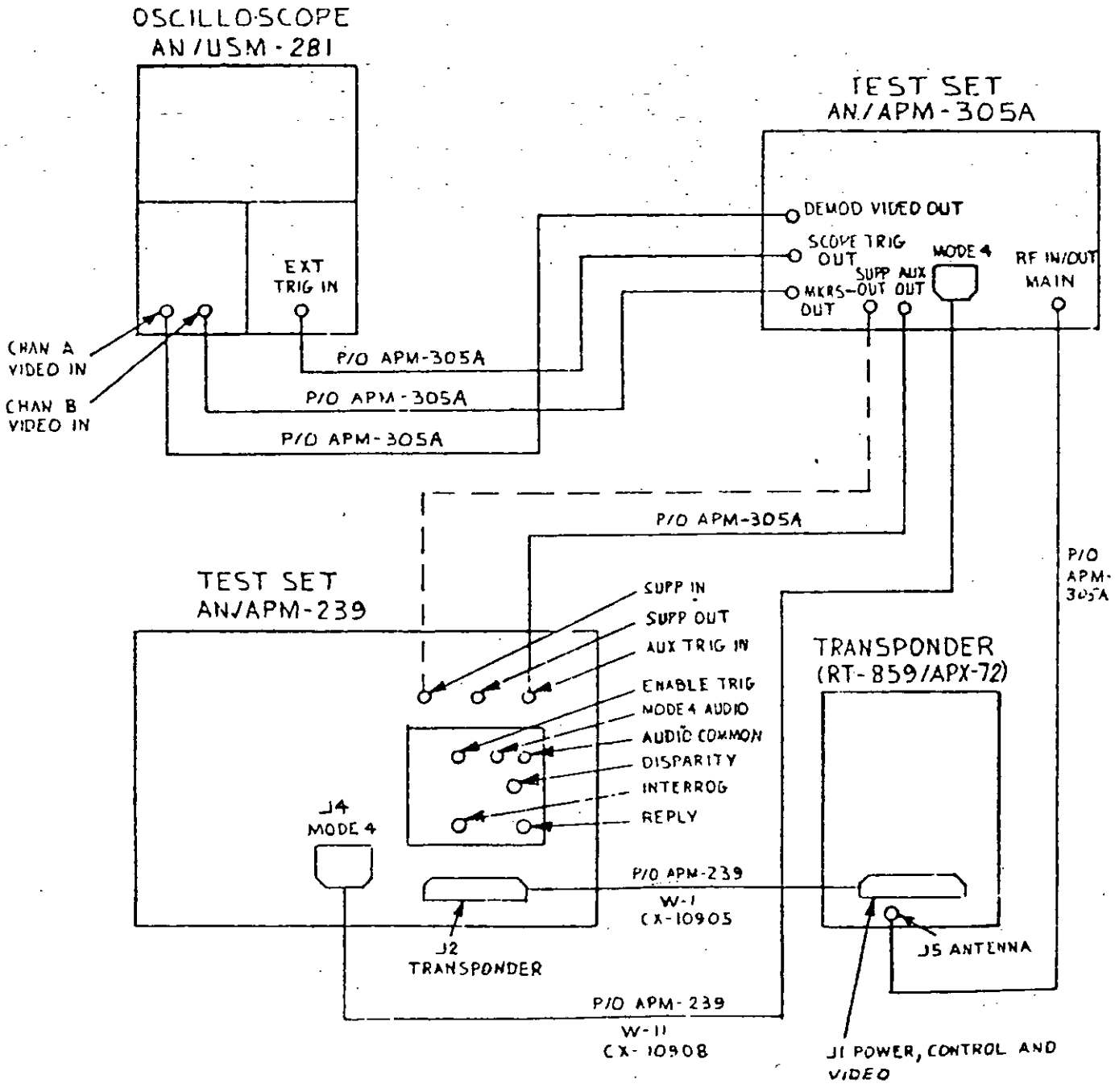



Figure 1



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4.15.5 SHOCK. (See 3.18.5).

4.15.5.1 BENCH HANDLING. The equipment shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure V. The performance tests before and after shall be conducted as specified in Table IV (See 3.18.5.1).

4.15.5.2 BASIC DESIGN. The equipment, with cover attached, shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure 1. The shock pulse characteristics shall be in accordance with Figure 516.1-2, except that the amplitude shall be 15 g's and the time duration 11 milliseconds. The performance tests before and after shall be conducted as specified in Table IV (See 3.18.5.2).

4.15.6 FUNGUS. The unit, with its cover separated from the equipment, shall be tested in accordance with Method 508, Procedure 1 of MIL-STD-810. No cleaning of the unit(s) is permitted for 72 hours prior to the fungus test. Handling prior to and during testing shall be accomplished without contamination of the equipment. The interior of the case shall also be sprayed and the unit(s) shall be subjected to the test in an open condition. There shall be abundant growth colonization on 50 percent or more of the area of the control items after 14 and 28 days of test. At the conclusion of the incubation period, the unit(s) shall be inspected and show no more than 0.015 square inch in area, of sparse microbial growth as evidenced by growth colonization (which includes branching and sporulation) on or within each cubic foot or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation (See 3.18.6 and 6.5).

4.15.7 SALT FOG. The unit, with its cover separated from the equipment, shall be subjected to the salt fog test in accordance with Method 509 of MIL-STD-810 (See 3.18.7).

4.16 ELECTRICAL PERFORMANCE TESTS. (See 3.19).

4.16.1 PULSE REPETITION FREQUENCY (PRF). (See 3.19.1)

4.16.1.1 PRF SYNCHRONIZATION. With the use of a dual-trace oscilloscope, determine that a single Mode 3/A challenge video signal (3.19.4.3.3) is generated synchronously with each PRF cycle (See 3.19.1.1).



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TABLE IV

| | Low Temp | | High Temp | | Alt. | | Humi-dity | | Vib. | | Shock Bench | | Shock Basic | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Req. Para. | Before/After | During | Before/After | During | Before/After | During | Before/After | After | Before/After | During | Before/After | During | Before/After | During |
| 3.19.1.1.1.1 | X | X | X | X | | | | | X | | | | X | |
| 3.19.1.1.1.2 | | | | | | | | | X | | | | X | |
| 3.19.1.1.2 | X | X | X | X | | | X | X | X | | | | X | |
| 3.19.4.2 | X | X | X | X | | | | | X | | X | | X | |
| 3.19.4.3.3 | X | X | X | X | | | X | X | X | | X | | X | |
| 3.19.4.3.6 | X | X | X | X | X | X | X | X | X | | X | | X | |
| 3.19.4.3.6.1 | X | X | X | X | | | | | X | | X | | X | |
| 3.19.4.3.8 | X | X | X | X | | | | | X | | | | X | |
| 3.19.4.4.2 | | | | | | | | | X | | | | X | |
| 3.19.4.5.2 | | | | | | | | | X | | | | X | |
| 3.19.4.6.1.1 | X | X | X | X | | | X | X | X | | | | X | |
| 3.19.4.6.1.2 | X | X | X | X | | | X | X | X | | | | X | |
| 3.19.4.6.2.1 | X | X | X | X | X | X | X | X | X | | | | X | |
| 3.19.4.6.2.2 | X | X | X | X | X | X | X | X | X | | | | X | |
| 3.19.4.6.4.2 | | | | | | | | | X | | | | X | |
| 3.19.5 | X | X | X | X | | | | | | | | | | |
| 3.19.6.2 | X | X | X | X | | | X | X | | | | | | |
| 3.19.6.4 | X | X | X | X | | | | | | | | | | |
| 3.19.7.1.1 | X | X | X | X | | | X | X | | | | | | |
| 3.19.7.1.1.3 | X | X | X | X | | | X | X | X | | | | X | |
| 3.19.7.2.3 | X | X | X | X | | | X | X | | | | | | |
| 3.19.7.2.4 | X | X | X | X | | | X | X | X | | | | X | |
| 3.19.8.1 | X | X | X | X | | | X | X | | | | | | |
| 3.19.8.2.1.1 | X | X | X | X | | | X | X | | | | | | |
| 3.19.8.2.1.2 | X | X | X | X | | | X | X | | | | | | |
| 3.19.10.6.1 | X | X | X | X | | | X | X | | | | | | |
| 3.19.10.6.2 | X | X | X | X | | | X | X | | | | | | |
| 3.19.10.8.5 | X | X | X | X | | | X | X | | | | | | |
| 3.19.11.1.1 | X | X | X | X | | | X | X | | | | | | |
| 3.19.11.2 | X | X | X | X | | | X | X | | | | | | |
| 3.19.11.3 | X | X | X | X | | | X | X | | | | | | |
| 3.19.11.4 | X | X | X | X | | | X | X | | | | | | |
| 3.19.11.7.1.1 | | | | | | | | | X | | X | | X | |
| 3.19.11.7.1.2 | | | | | X | X | | | X | | X | | X | |
| 3.19.11.7.2 | | | | | | | | | X | | X | | X | |



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4.16.1.1.1 FIXED FREQUENCY MODE. (See 3.19.1.1.1).

4.16.1.1.1.1 BASIC PRF. While varying the PRF front Panel controls from Minimum setting, measure the PRF (See 3.19.1.1.1.1).

4.16.1.1.1.2 REDUCED PRF. Using the test of 4.16.1.1.1.1, verify that the equipment has a front panel switch which reduces by one-half the basic PRF range (See 3.19.1.1.1.2).

4.16.1.1.2 SWEEP FREQUENCY MODE. Set the equipment controls for sweep frequency mode of operation. Measure the repetition rate of the sweep sync trigger (3.19.3.1.3) with the equipment operating in the narrow sweep and wide sweep modes of operation (See 3.19.1.1.2).

4.16.2 "0" (ZERO) TRIGGER. Using an oscilloscope, verify that an internal trigger is generated and occurs prior to the generation of the equipment's challenge video signals. Determine that the challenge video signals are generated synchronously with the "0" trigger (See 3.19.2).

4.16.3 TRIGGER OUTPUT. (See 3.19.3).

4.16.3.1 SCOPE TRIGGER. (See 3.19.3.1).

4.16.3.1.1 EXTERNAL VARIABLE DELAY TRIGGER. With the equipment in the fixed-frequency mode of operation, measure the characteristics of the output trigger. Measure the delay between the external variable delay trigger and the "0" trigger for minimum setting of the delay trigger's front panel controls (See 3.19.3.1.1).

4.16.3.1.1.1 JITTER. Measure the jitter of the delay trigger with respect to the "0" trigger for a delay setting of 4000 USEC (See 3.19.3.1.1.1).

4.16.3.1.2 FIXED TRIGGER - DUAL 3/A. Set the equipment controls to provide Dual 3/A challenge video signals. Measure the characteristics of the output trigger. Measure the output trigger timing with respect to the "0" trigger (See 3.19.3.1.2).

4.16.3.1.3 SWEEP SYNC TRIGGER. Set the equipment controls for sweep frequency operation. Measure the characteristics of the output trigger. Measure the output trigger timing with respect to the start of each sweep cycle (See 3.19.3.1.3).

4.16.3.2 AUXILIARY TRIGGER. (See 3.19.3.2).



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4.16.3.2.1 AUXILIARY TRIGGER IN SWEEP FREQUENCY. Set the equipment controls for sweep frequency operation. Set the auxiliary trigger "ON-OFF" switch to "ON". Measure the characteristics (3.19.3.2) of the auxiliary output trigger. Measure the timing of the auxiliary trigger with respect to the external variable delay trigger. Verify that the output disappears when the switch is set to "OFF" (See 3.19.3.2.1).

4.16.3.2.2 AUXILIARY TRIGGER IN FIXED FREQUENCY. Set the equipment controls for fixed frequency operation. Measure the characteristic (3.19.3.2) of the output auxiliary trigger. Measure the timing of the auxiliary trigger with respect to the "0" trigger. Verify that the output disappears when the switch is set to "OFF" (See 3.19.3.2.2).

4.16.4 CHALLENGE VIDEO. Challenge video signals are not available from a front panel output, therefore, tests 4.16.4.1 thru 4.16.4.6.4.2 shall be performed using RF signals demodulated by the standard biased detector shown in Figure 2 (See 3.19.4).

4.16.4.1 FIXED FREQUENCY. Verify that challenge video signals are generated synchronously with the "0" trigger (See 3.19.4.1).

4.16.4.2 SWEEP FREQUENCY. Set the equipment controls for sweep frequency operation. Verify that the challenge video signals are generated synchronously with the external variable delay trigger (See 3.19.4.2).

4.16.4.2.1 CHALLENGE VIDEO DELAY CONTROL. Set the equipment controls for sweep frequency operation. Measure the spacing between the sweep sync trigger and the challenge video pulses while varying the external variable delay trigger control from its minimum to maximum settings (See 3.19.4.2.1).

4.16.4.3 CHALLENGE VIDEO SELECT. (See 3.19.4.3).

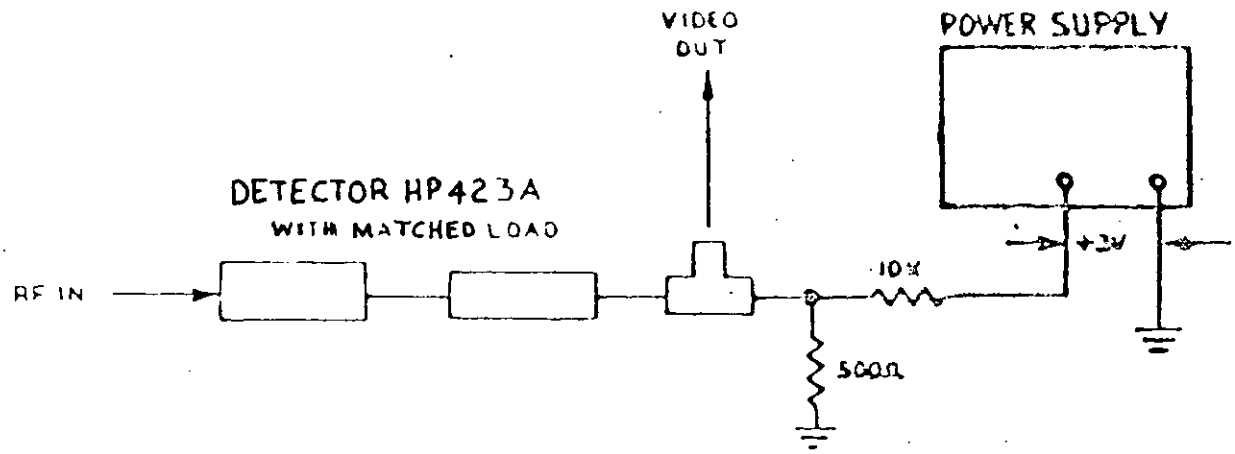
4.16.4.3.1 CHALLENGE - MODE 1. Set the challenge video select control for Mode 1. Measure the spacing between the two challenge pulses (See 3.19.4.3.1).

4.16.4.3.2 CHALLENGE - MODE 2. Set the challenge video select control for Mode 2. Measure the spacing between the two challenge pulses (See 3.19.4.3.2).

4.16.4.3.3 CHALLENGE - MODE 3/A. Set the challenge video select control for Mode 3/A. Measure the spacing between the two challenge pulses (See 3.19.4.3.3).



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Standard Biased Detector Network

Figure 2



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4.16.4.3.4 CHALLENGE - MODE C. Set the challenge video select control for Mode C. Measure the spacing between the two challenge pulses (See 3.19.4.3.4).

4.16.4.3.5 CHALLENGE - TEST MODE. Set the challenge video select control for Test Mode. Measure the spacing between the two challenge pulses (See 3.19.4.3.5).

4.16.4.3.6 CHALLENGE - MODE 4(WORD A). Set the challenge video select control for Mode 4(Word A). Verify that twenty-eight (28) pulses are generated (See 3.19.4.3.6).

4.16.4.3.6.1 CHALLENGE - MODE 4 SPACING ACCURACY. Measure the position accuracy of each pulse with respect to its nominal value (See 3.19.4.3.6.1).

4.16.4.3.7 CHALLENGE - MODE 4(WORD B). Set the challenge video select control for Mode 4(Word B). Verify that twenty-eight (28) pulses are generated (See 3.19.4.3.7).

4.16.4.3.7.1 CHALLENGE - MODE 4 SPACING ACCURACY. Measure the position accuracy of each pulse with respect to its nominal value (See 3.19.4.3.7.1).

4.16.4.3.8 CHALLENGE - DUAL 3/A. Set the challenge video select control for Dual 3/A. Measure the spacing between the two challenge pulses of each pulse pair (See 3.19.4.3.8).

4.16.4.3.8.1 CHALLENGE - DUAL 3/A VARIABLE SPACING. Measure the position of the second challenge pulse pair with respect to the first challenge pulse pair while varying the external variable delay trigger control from its minimum to its maximum setting (See 3.19.4.3.8.1).

4.16.4.3.9 CONTINUOUS WAVES(CW). Set the challenge video select control to CW. Measure to verify that the power level at each of the equipment's RF outputs (3.19.10) is -10 dBm $\pm$1 dB using a CW RF power meter (See 3.19.4.3.9).

4.16.4.4 VARIABLE CHALLENGE PULSE SPACING. (See 3.19.4.4).

4.16.4.4.1 SPACING SELECT. For challenge video select control settings of Mode 1, 2, 3/A, Test, C, and Dual 3/A, measure the amount and accuracy of the selectable spacings for the challenge pulse pairs (P1, P3). For challenge video select control settings of Mode 4 (Word A) and Mode 4(Word B), measure the amount and accuracy of the selectable spacings for each of the second, third and fourth challenge pulses with respect to the first challenge pulse (See 3.19.4.4.1).



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4.16.4.4.2 CONTINUOUS SPACING. Measure the amount of spacing deviation, selected by front panel controls, of each of the movable challenge pulses about their nominal positions (See 3.19.4.4.2).

4.16.4.5 CHALLENGE VIDEO PULSE WIDTH. (See 3.19.4.5).

4.16.4.5.1 WIDTH SELECT. For the challenge video signals of 3.19.4.3.1 through 3.19.4.3.8, Measure the amount and accuracy of the selectable width settings provided by front panel controls for the challenge pulses (See 3.19.4.5.1).

4.16.4.5.2 CONTINOUSLY VARIABLE WIDTH. Measure the Mode 3/A challenge pulse width with the continuous width control set to its minimum and maximum settings (See 3.19.4.5.2).

4.16.4.6 ISLS CONTROL PULSE(P2). (See 3.19.4.6).

4.16.4.6.1 ISLS - MODES 1, 2, 3/A, TEST C, AND DUAL 3/A.. (See 3.19.4.6.1).

4.16.4.6.1.1 NOMINAL SPACING. Set the equipment controls for ISLS operation. For challenge Video select control settings of Mode 1, 2, 3/A, Test, C, and Dual 3/A, measure the spacing of the ISLS pulse with respect to the first challenge pulse (P1) (See 3.19.4.6.1.1).

4.16.4.6.1.2 PULSE WIDTH. Measure the width of the ISLS pulse (See 3.19.4.6.1.2).

4.16.4.6.2 ISLS - MODE 4(WORDS A AND B). (See 3.19.4.6.2).

4.16.4.6.2.1 NOMINAL SPACING. Set the equipment controls for ISLS operation. For challenge video select control settings of Mode 4(Word A) and Mode 4(Word B), measure the spacing of the ISLS pulse with respect to the first challenge pulses (See 3.19.4.6.2.1).

4.16.4.6.2.2 PULSE WIDTH. Measure the width of the ISLS pulse (See 3.19.4.6.2.2).

4.16.4.6.3 ISLS - DUAL 3/A MODE. Set the equipment controls for ISLS operation. For a challenge video select control settings of Dual 3/A, verify that the ISLS pulse is missing from the second challenge pulse pair using the test of 4.16.4.6.1.1 (See 3.19.4.6.3).

4.16.4.6.4 VARIABLE ISLS PULSE SPACING. (See 3.19.4.6.4).

4.16.4.6.4.1 SPACING SELECT. Set the equipment controls for ISLS operation. For the challenge video signals of 3.19.4.3.1.



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through 3.19.4.3.8., measure the amount and accuracy of the selectable ISLS spacings provided by the front panel controls (See 3.19.4.6.4.1).

4.16.4.6.4.2 CONTINUOUS SPACING. Measure the amount of spacing deviation, selected by the front panel controls, of the ISLS pulse about its nominal position for Mode 3/A challenges (See 3.19.4.6.4.2).

4.16.5 SUPPRESSION PULSE OUTPUT. Set the suppression pulse output switch to "ON". Measure the characteristics of the suppression output pulse. Verify that the output disappears when the switch is set to "OFF" (See 3.19.5).

4.16.6 MODE 4. (See 3.19.6).

4.16.6.1 MODE 4 REPLY OUTPUT. Set the equipment controls for Mode 4 operation. Using a pulse generator, apply a simulated Mode 4 enable trigger (3.19.6.1) to the equipment through the Mode 4 interface connector (3.19.6.5). Measure the characteristics of the Mode 4 reply (See 3.19.6.2).

4.16.6.2 MODE 4 DISPARITY PULSE CHARACTERISTICS. Set the equipment controls for Mode 4 ISLS operation. Verify that the Mode 4 reply is inhibited. Measure the characteristics of the generated disparity pulse (3.19.6.4). Set the Inhibit Control Switch (3.19.6.4.1) to the disparity position. Verify that the Mode 4 reply output is inhibited. Measure the characteristics of the generated disparity pulse (See 3.19.6.4).

4.16.6.3 MODE 4 INTERFACE CONNECTOR. Inspect the equipment connector to verify its type number (See 3.19.6.5).

4.16.7 MARKERS. (See 3.19.7).

4.16.7.1 FIXED FREQUENCY. (See 3.19.7.1).

4.16.7.1.1 SELECTIVE IDENTIFICATION FEATURE (SIF) REPLY MARKERS. Verify the number and timing characteristics of the output markers (See 3.19.7.1.1).

4.16.7.1.1.1 MARKER PHASING. Verify that the timing markers vary simultaneously when the equipment front panel phasing control is set to its minimum and maximum settings (See 3.19.7.1.1.1).

4.16.7.1.1.2 INITIAL MARKER TIMING. For challenge video select control settings of Mode 1, 2, 3/A, Test, C, and Dual 3/A, measure the timing between the occurrence of the last challenge pulse (P3) and the occurrence of the initial SIF timing marker (See 3.19.7.1.1.2).



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4.16.7.1.1.3 CHARACTERISTICS. Measure the marker characteristics (See 3.19.7.1.1.3).

4.16.7.1.2 TIMING MARKERS. Verify that the markers are generated synchronously with the challenge video (3.19.4) and are provided at a front panel connector (See 3.19.7.1.2).

4.16.7.1.2.1 MARKER SPACING. Measure the marker spacing and accuracy (See 3.19.7.1.2.1).

4.16.7.1.2.2 CHARACTERISTICS. Measure the marker characteristics (See 3.19.7.1.2.2).

4.16.7.2 SWEEP FREQUENCY. Set the equipment controls for sweep frequency operation. Verify that frequency marker outputs are provided during RF sweep operation (See 3.19.7.2).

4.16.7.2.1 NARROW SWEEP BANDWIDTH. Set the equipment controls for ±5 MHz sweep frequency operation. Measure the number of markers (See 3.19.7.2.1).

4.16.7.2.2 WIDE SWEEP BANDWIDTH. Set the equipment controls for ±20 MHz Sweep Frequency operation. Measure the number of markers (See 3.19.7.2.2).

4.16.7.2.3 MARKER ACCURACY. Measure the accuracy of the frequency markers (See 3.19.7.2.3).

4.16.7.2.4 CHARACTERISTICS. Measure the characteristics of the frequency marker outputs (See 3.19.7.2.4).

4.16.8 RF SIGNAL GENERATION. (See 3.19.8).

4.16.8.1 FIXED FREQUENCY. Measure the frequency of the equipment RF outputs at one of the input/output connectors (3.19.10.2) (See 3.19.8.1).

4.16.8.2 SWEEP FREQUENCY. (See 3.19.8.2).

4.16.8.2.1 SWEEP BANDWIDTH. (See 3.19.8.2.1).

4.16.8.2.1.1 NARROW BAND. Set the equipment controls for ±5 MHz sweep frequency operation. Measure the sweep bandwidth of the equipment about the nominal center frequency of 1030 MHz (See 3.19.8.2.1.1).



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4.16.8.2.1.2 WIDE BAND. Set the equipment controls for ±20 MHz sweep frequency operation. Measure the sweep bandwidth of the equipment about the nominal center frequency of 1030 MHz (See 3.19.8.2.1.2).

4.16.8.2.1.3 SWEEP BLANKING. Set the equipment controls for sweep frequency operation. During the sweep off time, verify that there is no modulated RF output from either of the equipment's output channels (See 3.19.8.2.1.3).

4.16.9 LOW LEVEL RF INPUT Verify low level RF input characteristics while conducting test 4.16.11.1.2 (See 3.19.9).

4.16.10 HIGH LEVEL RF INPUT/OUTPUT. (See 3.19.10).

4.16.10.1 INPUT/OUTPUT VSWR. Measure the VSWR of both input/output channels at the connectors (See 3.19.10.3).

4.16.10.2 ATTENUATION. Verify that the RF output level at each input/output connector (3.19.10.2) can be varried between -10 dBm and -90 dBm (See 3.19.10.4).

4.16.10.2.1 ATTENUATOR SCALES. Verify the attenuator calibration data to the attenuator markings for both the MAIN and AUXILIARY attenuators to ascertain that the attenuator is calibrated as specified (See 3.19.10.4.1).

4.16.10.2.2 DIFFERENTIAL ATTENUATION. Using the attenuator calibration data, verify the differential attenuation between every two consecutive 10 dBm scale divisions (See 3.19.10.4.2) on the MAIN and AUXILIARY attenuators (See 3.19.10.4.3).

4.16.10.3 ADJUSTABLE INDICATOR. Set the AUXILIARY attenuator to settings of -10 dBm, -11 dBm and -12 dBm. For each setting, verify that the adjustable indicator can be set to 0 dB on the ISLS scale (See 3.19.10.5.2).

4.16.10.4 RF OUTPUT LEVEL ACCURACY. (See 3.19.10.6).

4.16.10.4.1 FIXED FREQUENCY. Set the pulse width control to 0.80 USEC. For MAIN attenuator settings of -10 dBm, -15 dBm and -20 dBm measure the equipment's RF output level at the front panel output connector RF input/output MAIN (3.19.10.2). Repeat the test using AUXILIARY attenuator settings (See 3.19.10.6.1).

4.16.10.4.2 SWEEP FREQUENCY. Set the equipment controls for ±5 MHz sweep frequency operation and CW modulation. Measure the amplitude



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variation in the sweep output level over the ±5 MHz bandwidth for MAIN and AUXILIARY channel outputs. Repeat the above test for the ±20 MHz sweep bandwidth (See 3.19.10.6.2).

4.16.10.4.2.1 SWEEP OFFSET. While performing the test of 4.16.10.4.1, record the equipment's MAIN output level for an attenuator setting of -10dBm. Without resetting the MAIN attenuator, perform the test of 4.16.10.4.2 and verify that the sweep output level at center frequency (1030 MHz) is within 0.5 dB of the RF output level established for the fixed frequency test. Repeat for AUXILIARY (See 3.19.10.6.2.1).

4.16.10.5 ISLS RF OUTPUT. Set the equipment controls for ISLS operation. Verify that the equipment RF output is from the MAIN input/output connector and consists of the two challenge pulses (P1 and P3) and the ISLS pulse (P2) (See 3.19.10.7).

4.16.10.5.1 AUXILIARY LEVEL. Set the equipment controls for ISLS operation. Verify that the MAIN attenuator varies the level of the combined challenge video signal (3.19.10.7.1) and the AUXILIARY attenuator varies the ISLS pulse amplitude with the composite video signal (3.19.10.7.2). Set the MAIN and AUXILIARY attenuators to -10dBm and measure the amplitudes of the challenge pulses and the ISLS pulse (3.19.10.7.3) level set accuracy. Decrease the AUXILIARY attenuator setting to -19dBm, verify that the ISLS amplitude decreased 9 +0.5 dB with respect to the original ISLS pulse amplitude.

4.16.10.6 MODULATION. Set the equipment front panel controls (3.19.10.8.1) to provide the challenge video signals specified in 3.19.4.3.1 to 3.19.4.3.9. For each challenge video signal selected, observe the corresponding pulse-modulated RF output signal from the equipment's MAIN and AUXILIARY outputs (See 3.19.10.8).

4.16.10.6.1 CW LEAKAGE. Set the MAIN attenuator setting to -10 dBm and the AUXILIARY attenuator setting to -90 dBm. Measure the amplitude of the CW leakage signal coming from the AUXILIARY channel output. Repeat for the MAIN channel output with the MAIN attenuator setting to -90 dBm and the AUXILIARY attenuator setting to -10 dBm (See 3.19.10.8.2).

4.16.10.6.2 MODULATOR ON-OFF RATIO. Set the MAIN and AUXILIARY attenuators to -10 dBm. Measure the on-to-off ratios of the equipment's modulated RF output signals (See 3.19.10.8.3).

4.16.10.6.3 AUXILIARY CHANNEL DELAY RANGE. Measure the demodulated pulse characteristics of equipment's MAIN and AUXILIARY outputs simultaneously on a dual-trace oscilloscope. Measure the delay between the MAIN video signal and the AUXILIARY video signal while



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the front panel delay range control is varied from its minimum to its maximum settings (See 3.19.10.8.4).

4.16.10.6.3.1 AUXILIARY CHANNEL DELAY CONTROL DETENT. Verify that a detent position is provided in the delay control at the specified delay setting (See 3.19.10.8.4.1).

4.16.10.6.4 MODULATION FIDELITY. Measure the characteristics of the equipment's Mode 3/A and Mode 4 (Word A) modulated RF output signals while performing the test of 4.16.10.6 (See 3.19.10.8.5).

4.16.11 MEASUREMENT CAPABILITY. The measurement capability of the equipment shall be performed utilizing the AN/USM-281 Oscilloscope or equivalent (See 3.19.11).

4.16.11.1 DEMODULATOR SENSITIVITY. (See 3.19.11.1).

4.16.11.1.1 HIGH LEVEL. Apply a pulse-modulated RF signal, having the following characteristics, to the MAIN channel input connector:

| | |
|---|---|
| Pulse Width | 0.45 ±0.1 USEC |
| PRF | 1000 Hz ±5% |
| Power | +18 dBw(±0 dB, -0.5 dB) |
| | +33 dBw(+0.5 dB, -0 dB) |

For input frequencies of 1085 ±0.1 MHz, 1090 ±0.1 MHz and 1095 ±0.1 MHz, measure the amplitude and PRF of the demodulated video output. Repeat the above measurements for the AUXILIARY input channel (See 3.19.11.1.1).

4.16.11.1.2 LOW LEVEL. Repeat the test of 4.16.11.1.1 with an input signal of -12 dBm(+0 dBm, -0.5 dBm) applied to the low level input connector (3.19.9). Measure the characteristics of the demodulated video output. Repeat the test with an input signal of +3 dBm (+0.5 dB, -0 dB) (See 3.19.11.1.2).

4.16.11.2 PULSE FIDELITY. Apply a +18 dBw RF signal defined in 4.16.11.1.1, to the standard biased detector with appropriate attenuation presented in Figure 2. Measure the characteristics of the demodulated video output. Apply the same signal to each of the equipment's high power input channels and measure the characteristics of the demodulated video output. Compare the results. Repeat the above tests for an RF input power level of +33 dBw ±0.5 dB (See 3.19.11.2).

4.16.11.3 PULSE POWER MEASUREMENT. Apply a +18 dBw, 1085 ±0.1 MHz, RF signal with a pulse width of 0.45 ±0.1 USEC and a PRF of



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1000 Hz ±5% to each of the equipment's high power input channels. Measure the power level of the input signal. Repeat the above tests for the following input signals:

| | POWER LEVEL | FREQUENCY |
|---|---|---|
| a. | 20 dBw ±1.0 dB | 1085 ±0.1 MHz |
| b. | 26 dBw ±0.5 dB | 1090 ±0.1 MHz |
| c. | 29 dBw ±0.5 dB | 1090 ±0.1 MHz |
| d. | 33 dBw ±1.0 dB | 1095 ±0.1 MHz |
| e. | -12 dBm ±1.0 dB | 1085 ±0.1 MHz |
| f. | 0 dBm ±1.0 dB | 1090 ±0.1 MHz |
| g. | 3 dBm ±1.0 dB | 1095 ±0.1 MHz |
| h. | 30 dBm ±1.0 dB* | 1090 ±0.1 MHz |

*Plus accuracy of accessory attenuators

Apply input signals e, f, g and h above to the equipment's low power input connector using necessary attenuators (See 3.19.11.3 and 3.20.6).

4.16.11.3.1 POWER BIAS CONTROL. While performing the test of 4.16.11.3 verify that the demodulated video output is reducible to a 1 volt level for the input power levels specified (See 3.19.11.3.1).

4.16.11.4 FREQUENCY MEASUREMENT. Set the equipment controls for frequency measurement. Apply a pulse-modulated RF signal to the equipment's MAIN channel input having the following characteristics:

| | |
|---|---|
| Pulse Width | 0.45 ±0.1 USEC |
| PRF | 1000 Hz ±5% |
| Power | +18 dBw(±0 dB, -0.5 dB) |
| Frequency | 1070 MHz, 1087 MHz, 1090 MHz, 1093 MHz, 1110 MHz |

Measure the frequency of the input signals. Repeat the above measurements with an input power level of 33 dBw to the AUXILIARY input channel (See 3.19.11.4).



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4.16.11.5 PRF MEASUREMENT. Set the equipment controls for a PRF of 10 Hz. Using the equipment's measurement capabilities, measure the PRF of the equipment. Measure the equipment's PRF using a frequency counter and compare the measured results. Repeat the test for PRF's of 100 Hz, 1000 Hz and 10,000 Hz. While performing the test of 4.16.11.1.1 at the RF frequency of 1090 MHz ±0.1 MHz, measure the PRF of the applied RF signal using the equipment (See 3.19.11.5).

4.16.11.6 SELF TEST. (See 3.19.11.7).

4.16.11.6.1 SIGNAL ROUTING. Connect either of the equipment MAIN or AUXILIARY RF outputs to the low power input connector. perform the tests of 4.16.11.6.1.1 and 4.16.11.6.1.2 (See 3.19.11.7.1).

4.16.11.6.1.1 PULSE FIDELITY MEASUREMENT. Measure the pulse characteristics of the equipment's modulated RF outputs. Apply the equipment's RF outputs to the standard biased detector, Figure 2, and measure the pulse characteristics. Compare the results of the two measurement methods (See 3.19.11.7.1.1).

4.16.11.6.1.2 POWER MEASUREMENT ACCURACY. Using the test setup of 4.16.11.6.1, measure the equipment peak power output (See 3.19.11.7.1.2).

4.16.11.6.2 BUILT-IN-TEST (BIT) FOR FREQUENCY MEASUREMENT. Set the equipment controls for sweep frequency operation. Using the front panel BIT control and the equipment frequency measurement capability, measure the frequency of the equipment's 1090 MHz BIT RF source (See 3.19.11.7.2).

4.16.12 POWER SUPPLY FAULT INDICATOR. In accordance with the following table, vary, in turn, each of the working voltages and verify the indicator illumination. The +1.1 volt variation from nominal test will not be required for the +5 volt working voltage (See 3.19.12):

| VOLTAGE VARIATION FROM NOMINAL | INDICATOR ILLUMINATION |
|---|---|
| -1.1 | Yes |
| -0.2 | No |
| +0.2 | No |
| +1.1 | Yes |

4.16.13 POWER SUPPLY SHORT CIRCUIT PROTECTION. In turn, short circuit each of the equipment's DC working voltages for a period of one minute each. After the shorted condition is removed, verify



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that the supply voltage returns to its specified level and that no damage has been done (See 3.19.13).

4.17 ACCESSORIES. (See 3.20).

4.17.1 CABLE ASSEMBLIES. Verify DC continuity for the conductors in each cable assembly. Verify that a DC short circuit condition does not exist between conductors (See 3.20.1, 3.20.2, 3.20.3 and 3.20.4).

4.17.2 VIDEO TERMINATIONS. Visually inspect the terminations to determine that the appropriate connectors have been used. Measure the DC resistance of each termination (See 3.20.7).

4.17.3 FIXED ATTENUATORS. Terminate the 15 dB attenuator with a suitable 50 ohm termination. Measure attenuator's VSWR and input impedance at frequencies of 1070, 1090, and 1110 ±0.5 MHz. Remove termination and measure attenuation using the same frequencies. Repeat for the 30 dB attenuator (See 3.20.6).

4.18 OPERABILITY. The equipment shall be operated under standard test conditions (4.9). Conduct test 4.16.11.6.1.2 (See 3.22).

4.19 INTERCHANGEABILITY. Like units, assemblies, subassemblies and replaceable parts shall be checked mechanically and electrically to determine conformance to the interchangeability requirement (See 3.22).

4.20 VISUAL AND MECHANICAL INSPECTION. Assembled equipment shall be examined for defects listed in MIL-STD-252. Prior to assembly into the equipment, printed circuits boards shall be examined for compliance with MIL-P-55110. Assemblies, modules and sections shall be examined for compliance with MIL-STD-252 (See 3.24).

4.21 QUALITY CONFORMANCE INSPECTION FOR PACKAGING. Quality Conformance for Packaging shall be in accordance with Group A, Group B, and when required (See 6.2d), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING. Packing shall be in accordance with SPI 1G00119.

6. NOTES.

6.1 INTENDED USE. Test Set, Transponder Set AN/APM-305A will be specifically used to align, calibrate and troubleshoot transponder systems, such as the AN/APX-72 at direct and general support levels of maintenance. The equipment provides the necessary test stimuli for maintaining transponder systems in the field.



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6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. First Article inspection: Several samples of each item cited in Section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while electrical performance measurements can be made on other sample units (See 3.2).

c. Marking and shipping of samples.

d. Place of final inspection.

e. Submission of the statement of treatment referenced in MIL-P-11268 paragraph titled Toxicity as soon as possible after award of contract. This statement should be submitted to the Contracting Officer.

f. Quantity of spare parts required.

6.3 GROUP C INSPECTION. Approval to ship unit(s) on order may be withheld, at the discretion of the Government, pending decision from the procuring activity until compliance with 4.5.4 is completed.

6.4 LOCATION OF OPERATIONAL INSPECTION. It is desirable that the operational inspection 4.18 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected at the packaging site immediately prior to packaging.

6.5 DEFINITIONS.

6.5.1 BRANCHING. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem of filament (hypha).

6.5.2 GROWTH COLONIZATION. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hypae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.



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6.5.3 MICROBIAL GROWTH. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 SPORULATION. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.5.5 TUBULAR GERMINATION. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.6 DEVIATION. Request for deviation in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed designed change to withstand the applicable test for the selected method, including the rough handling test in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing material selected and economies realized.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-49056A (CR) | TEST SET, TRANSPONDER SET AN/APM- 305A |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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